Guidance on Writing System Requirements and Actions

DAEN 459 Fall 2025

Guidance for Teams on Writing System Requirements and Actions

In traditional projects that focus on system design or process improvement, system requirements specify how a tangible system will function in its future state. However, in research-focused or conceptual projects, teams may not be designing a physical system or implementing a process directly. Instead, they are exploring relationships, developing frameworks, or conducting analyses. Despite this difference, system requirements are still essential because they define what must be studied, modeled, or evaluated to ensure project success.

How Research-Focused Teams Should Write System Requirements

If your project is more research-oriented, think of system requirements as defining the scope of inquiry and the necessary components of your research framework rather than prescribing a tangible system's behavior. Here's how to structure them:

1. **Define what must be analyzed, modeled, or evaluated.**
   - Example: "The system shall include a model that simulates patient flow through the emergency department under varying demand scenarios."
2. **Specify the expected characteristics of deliverables (models, frameworks, etc.).**
   - Example: "The system shall incorporate a decision-support tool that ranks alternative scheduling policies based on efficiency and fairness criteria."

Ensure requirements remain verifiable and testable, even if qualitative.
Example: "The system shall provide a comparative analysis of at least three optimization techniques for workforce allocation in call centers." (Here, the requirement is met if the analysis is completed and structured as specified.)
Avoid vague or subjective language (e.g., "maximize efficiency" or "improve usability"). Instead, describe what will be developed or examined.
Instead of: "The system shall identify best practices in inventory management," **use:** "The system shall compile a taxonomy of inventory management strategies based on literature review and industry case studies."

Examples of Research-Oriented System Requirements

1. Supply Chain Analysis

Requirement: "The system shall include a model that quantifies the impact of lead time variability on supply chain performance using Monte Carlo simulation."

2. Healthcare Operations Research

Requirement: "The system shall compare at least three scheduling algorithms to evaluate their impact on patient wait times and provider utilization."

3. AI/Automation Study

Requirement: "The system shall analyze the performance of machine learning algorithms in detecting quality defects based on historical production data."

4. Sustainability Framework Development

Requirement: "The system shall establish an evaluation framework for measuring the environmental impact of alternative energy sources in manufacturing."

5. Human Factors & Ergonomics Research

Requirement: "The system shall conduct a workload analysis to assess the cognitive demand of different task assignments in an industrial assembly line."

Final Takeaway for Research Teams – System Requirements

While research projects may not focus on building a physical system, they still require structured system requirements to ensure clarity, consistency, and traceability. Instead of defining how a system must behave in operation, focus on defining what must be analyzed, compared, developed, or evaluated—keeping them clear, concise, and verifiable.
By approaching system requirements in this way, research teams can align their objectives with the capstone methodology while ensuring their work remains structured and impactful.

Clarifying Engineering Actions for Research-Focused Capstone Teams

In research-driven projects, system requirements often include elements that resemble actions because they define what must be studied, modeled, or evaluated rather than dictating how a physical system should behave. This can make it challenging to differentiate between system requirements and engineering actions. However, the key distinction is that system requirements specify *what* must be accomplished, while engineering actions define *how* those requirements will be fulfilled.

How Engineering Actions Differ from System Requirements

- **System Requirements (SRs):** Define what needs to be developed, studied, or analyzed. These ensure that the research outputs align with the project goals.

- **Engineering Actions (EAs):** Describe the specific tasks and methodologies your team will use to satisfy the system requirements. These outline the step-by-step work needed to generate research findings or conceptual deliverables.

Writing Engineering Actions for Research Projects

When writing engineering actions, research teams should focus on:

- Specific methodologies for addressing system requirements.
- Execution steps (e.g., data collection, modeling, case study analysis).
- Experimentation and validation efforts.
- Structuring research findings into usable deliverables.

Research-Oriented System Requirements and Engineering Actions

1. Supply Chain Analysis

System Requirement: "The system shall include a model that quantifies the impact of lead time variability on supply chain performance using Monte Carlo simulation."
Engineering Actions:

- Collect historical supply chain lead time data from industry sources or case studies.
- Develop a Monte Carlo simulation in Python or Excel to model variability.
- Perform sensitivity analysis to determine which lead time factors have the greatest impact on performance.
- Validate findings by comparing simulated outputs with real-world performance metrics.

2. Healthcare Operations Research

System Requirement: "The system shall compare at least three scheduling algorithms to evaluate their impact on patient wait times and provider utilization."
Engineering Actions:

- Identify three scheduling algorithms (e.g., first-come-first-serve, priority-based, machine learning-assisted).
- Develop a simulation model to test scheduling performance under varying patient demand conditions.
- Analyze results based on key performance indicators (e.g., average wait time, provider idle time).
- Summarize findings and provide recommendations on optimal scheduling practices.

3. AI/Automation Study

System Requirement: "The system shall analyze the performance of machine learning algorithms in detecting quality defects based on historical production data."
Engineering Actions:

- Obtain a dataset of historical production defect records.
- Train and test at least two machine learning models (e.g., decision trees, neural networks) for defect detection.
- Compare model accuracy, precision, and recall.
- Document insights and provide recommendations on algorithm selection for defect detection.

4. Sustainability Framework Development

System Requirement: "The system shall establish an evaluation framework for measuring the environmental impact of alternative energy sources in manufacturing."
Engineering Actions:

- Review existing literature on sustainability metrics for energy sources.
- Develop a scoring methodology to assess energy efficiency, carbon emissions, and cost trade-offs.
- Apply the framework to at least three alternative energy scenarios.
- Conduct expert interviews or case studies to validate framework effectiveness.

5. Human Factors & Ergonomics Research

System Requirement: "The system shall conduct a workload analysis to assess the cognitive demand of different task assignments in an industrial assembly line."
Engineering Actions:

- Identify key cognitive load measurement techniques (e.g., NASA-TLX, eye-tracking).
- Design and conduct experiments with participants performing various assembly tasks.
- Analyze cognitive workload data and compare across task types.
- Develop ergonomic recommendations for optimizing task distribution in assembly environments.

Key Takeaways for Research Teams

- System requirements define *what* must be analyzed, modeled, or evaluated.
- Engineering actions specify *how* your team will execute that research.
- System requirements may inherently include action-like phrasing, but engineering actions provide the step-by-step breakdown of how the requirement will be achieved.
- Well-defined engineering actions ensure that research projects remain structured, reproducible, and aligned with project goals.

By clearly distinguishing between system requirements and engineering actions, research teams can maintain a structured approach, ensuring their efforts lead to meaningful, validated, and actionable research outcomes.

Example Traceability Matrix

Customer Needs	System Requirements	Engineering Actions	Technical Objectives
N1: Improve scheduling efficiency	R1: The system shall evaluate scheduling algorithms based on efficiency and fairness.	EA1: Identify and implement at least three scheduling algorithms for comparison.	TO1: Develop a framework to assess trade-offs between scheduling policies.
N2: Reduce patient wait times	R2: The system shall simulate patient flow under various scheduling policies.	EA2: Develop and validate a discrete event simulation model of patient flow.	TO2: Provide decision support insights into optimal scheduling strategies.
N3: Optimize resource allocation	R3: The system shall analyze historical demand data to identify peak congestion periods.	EA3: Perform statistical analysis of historical patient demand data to detect patterns.	TO3: Identify key factors contributing to patient congestion and delays.